Homework 3

CALCULUS I

due: **Friday, February 13, 2026 at 11:30a**

Instructions: Please submit solutions on Populi as a knitted markdown pdf. For handwritten work, please scan your solution pages and include in the final pdf. Homeworks can be completed individually or in groups of up to three people. Names of all group members must be included on the writeups.

Problem 1: Derivatives in R with mosaicCalc

In this problem you will use the mosaicCalc package to define functions, compute their derivatives symbolically, and visualize them. The main functions you need are makeFun() to create a function from a formula and D() to compute a derivative. Example:

```
library(mosaicCalc)
f <- makeFun(x^3 - 4*x ~ x)
f(2)                    # evaluate f at 2
fp <- D(f(x) ~ x)       # derivative of f (creates a new function)
fp(2)                   # evaluate f'(2)
```

a. Define the function $f(x) = x^3 - 6x^2 + 9x + 1$ in R using makeFun(). Use D() to compute $f'(x)$ and $f''(x)$.

- Evaluate $f(0)$, $f'(0)$, and $f''(0)$.
- At which x is $f'(x) = 0$? (You can solve $3x^2 - 12x + 9 = 0$ by hand or use findZeros() from the mosaic package if you have it.)

b. Plot $f(x)$ and $f'(x)$ on the same set of axes over a domain that shows the main features (e.g., $x \in [0, 4]$). Label which curve is f and which is f'. Where does f have a local maximum or minimum? How does that relate to where $f'(x) = 0$?

c. Define $g(t) = 10t - t^2 + 0.1t^3$ (a position function in meters, t in seconds). Use D() to obtain the velocity function $v(t) = g'(t)$ and the acceleration function $a(t) = g''(t)$. Evaluate $v(2)$ and $a(2)$ and interpret them in one sentence each.

d. Use a central-difference approximation to check a derivative numerically. For $h(x) = \ln(x)$, the derivative is $h'(x) = 1/x$. Approximate $h'(2)$ using the central difference $\frac{h(2+\delta)-h(2-\delta)}{2\delta}$ with $\delta = 0.01$. Compare your result to the exact value $1/2 = 0.5$.

Problem 2: Application — Position, velocity, and acceleration

A particle moves along a line. Its position (in meters) at time t (in seconds) is given by

$$x(t) = t^3 - 12t + 5, \qquad t \geq 0.$$

a. Find the velocity function $v(t) = x'(t)$ and the acceleration function $a(t) = x''(t)$ (by hand or using your derivative rules).

b. At what time(s) is the particle at rest (velocity zero)? What is the position at those times?

c. At $t = 1$, is the particle speeding up or slowing down? Justify using the signs of $v(1)$ and $a(1)$.

d. Optional: In `R`, plot $x(t)$, $v(t)$, and $a(t)$ on $[0, 5]$ and briefly comment on how they relate.

Problem 3: Second derivative and concavity in `R`

a. Consider $f(x) = x^3 - 3x$ on the interval $[-2, 2]$.

- In `R`, define f with `makeFun()`, then compute $f'(x)$ and $f''(x)$ using `D()`.
- Plot $f(x)$ and $f''(x)$ on the same axes over $[-2, 2]$. Where is $f''(x) > 0$? Where is $f''(x) < 0$?
- State the intervals where f is concave up and where f is concave down. Where does the concavity change (inflection point)?

b. Consider $p(x) = e^{-x^2}$ (the Gaussian/hump function).

- Compute $p'(x)$ and $p''(x)$ in `R` using `D()`. Plot $p(x)$ and $p''(x)$ over $[-2, 2]$.
- Using the graph of $p''(x)$, identify (approximately) the intervals where p is concave up and where p is concave down. Where is the inflection point?

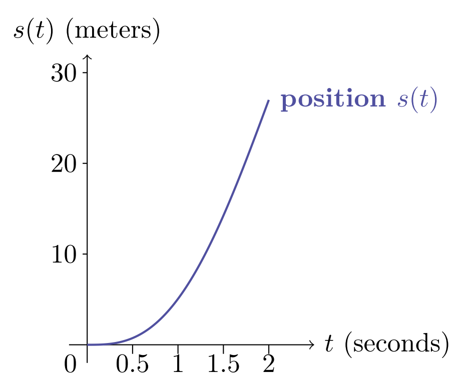
Problem 4: Jerk — Tesla Model S Plaid

The Tesla Model S Plaid is one of the quickest production cars in the world, capable of going from 0 to 60 mph in about 2 seconds. When you slam the accelerator, you don't just feel acceleration; you also feel how quickly that acceleration changes. That rate of change of acceleration is called **jerk**: $j(t) = a'(t) = s'''(t)$, the third derivative of position.

A simplified model for the car's position (in meters) from a standing start over the first 2 seconds is

$$s(t) = 6.75\,t^3 - 1.6875\,t^4, \qquad 0 \le t \le 2.$$

The graph below shows position $s(t)$ versus time t (in seconds) for this launch.



a. Find the velocity $v(t) = s'(t)$, acceleration $a(t) = s''(t)$, and jerk $j(t) = s'''(t)$. Give formulas and state their units (assuming t is in seconds and s in meters).

b. Evaluate $v(1)$, $a(1)$, and $j(1)$. At $t = 1$ second, is the car speeding up or slowing down? Is the acceleration increasing or decreasing? Explain briefly using the signs of $a(1)$ and $j(1)$.

c. At what time (in $(0, 2)$) does jerk equal zero? What is happening to the acceleration at that moment?

d. In `R` with `mosaicCalc`, define $s(t)$ using `makeFun()`, then use `D()` to compute v, a, and j. Plot all four on $[0, 2]$ and comment on how they relate (e.g., where jerk is zero and how that shows up on the acceleration curve).

Problem 5: Tangent lines and linear approximation

a. Let $f(x) = \sqrt{x + 1}$. Find the equation of the tangent line to the graph of f at $x = 3$. (You may use derivative rules or `mosaicCalc`; show your work.)

b. Use your tangent line from part (a) to approximate $\sqrt{4.1}$. Is your approximation an overestimate or underestimate? Briefly explain using the concavity of f.

c. In `R`, define $f(x) = \sqrt{x + 1}$ with `makeFun()`, compute $f'(x)$ with `D()`, and plot $f(x)$ together with its tangent line at $x = 3$ over an interval that includes $x = 3$ and $x = 3.1$. Comment on how well the tangent line approximates the curve near $x = 3$.